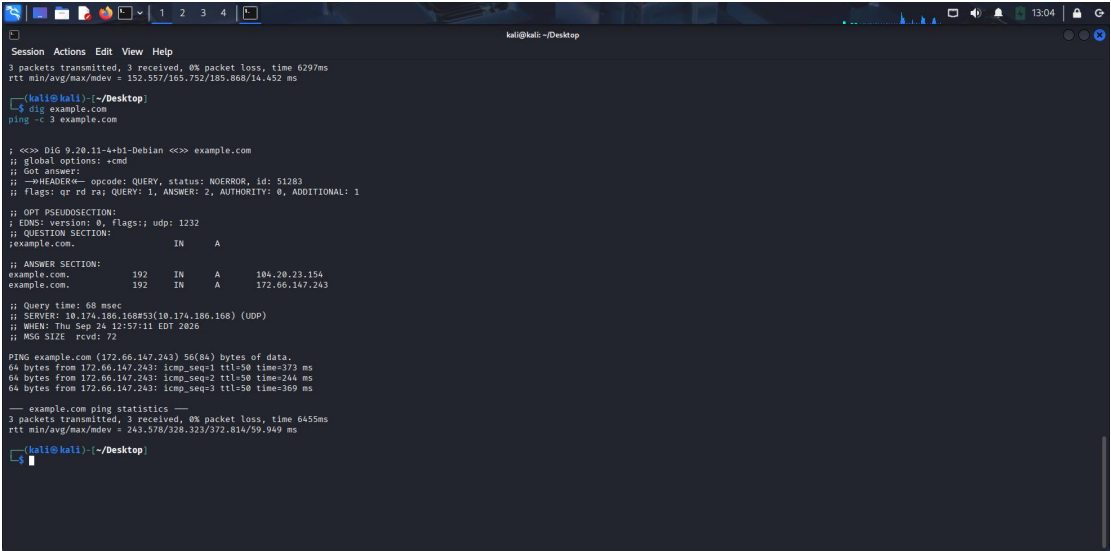
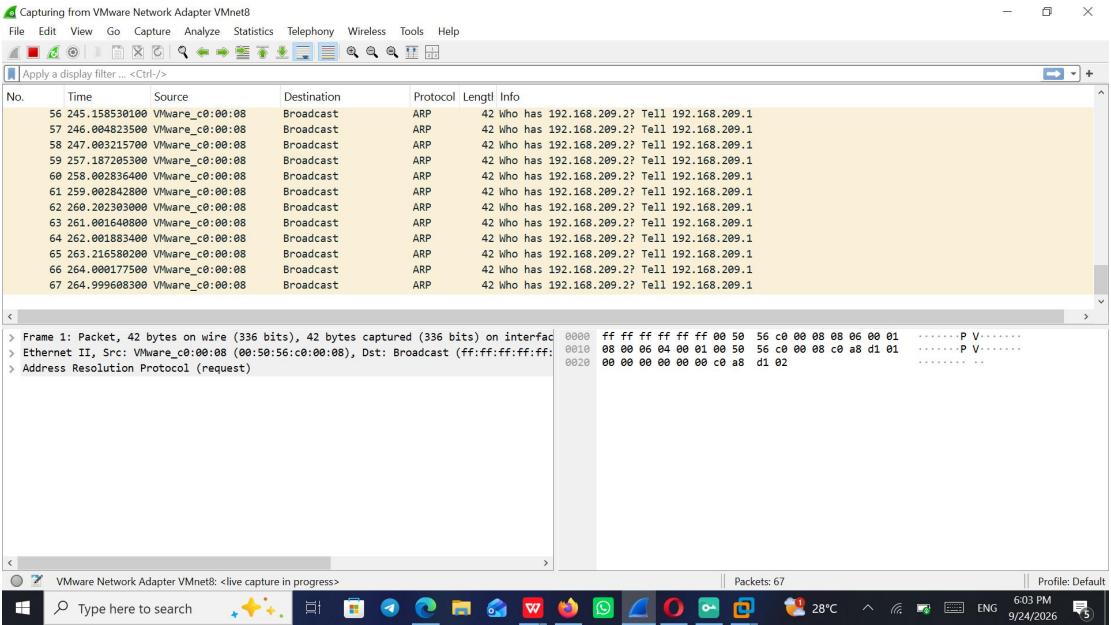


Name	Imoru Mayowa Peter
Student ID/Registration No	202521396FD
Course Code	Lab 9_SBT_DF 203
Lab Title	WEP40 Wireless Packet Decryption and Aircrack Forensics
Recommended Environment	Windows 10/11 host, Kali Linux and Wireshark +Aircrack
Submission Format	One PDF report plus the created files and-output evidence files
Course Code	SBT_DF 203
Instrutor name	Aminu Idris AMCPN
Date	25 September, 2026

EVIDENEC ONE



REPORT

Evidence 1 establishes the baseline. Wireshark shows normal ARP traffic, while the Kali terminal confirms legitimate DNS resolution and connectivity to example.com. No spoofing or interception is present at this stage.

EVIDENCE TWO

Original file hash (file.xz)/ Working copy hash (working_file.xz)

```
kali@kali: ~/Desktop
Session Actions Edit View Help
(kali@kali)~/Desktop
$ cd ~/Desktop

(kali@kali)~/Desktop
$ sha256sum file.xz
dd54144cae34f228bf4b87a9101ca71739693767ed357bd868861d4fab8d6  file.xz

(kali@kali)~/Desktop
$ cp file.xz working_file.xz
sha256sum working_file.xz
dd54144cae34f228bf4b87a9101ca71739693767ed357bd868861d4fab8d6  working_file.xz

(kali@kali)~/Desktop
$ unxz working_file.xz

(kali@kali)~/Desktop
$ sha256sum working_file.pcap
sha256sum: working_file.pcap: No such file or directory

(kali@kali)~/Desktop
$ ls -l ~/Desktop
-rw-r--r-- 1 kali kali 12208 220M 7z 2 lab_img_recovered.jpg
-rw-r--r-- 1 kali kali 12208 7z 2 lab_img_strings.txt
-rw-r--r-- 1 kali kali 12208 7z lab2a_sys_summary.sh
-rw-r--r-- 1 kali kali 12208 7z lab2a_sys_summary.sh
-rw-r--r-- 1 kali kali 12208 7z lab2a
-rw-r--r-- 1 kali kali 12208 7z 'mkdir_Desktop_abaa'
-rw-r--r-- 1 kali kali 12208 7z ruleset.txt
-rw-r--r-- 1 kali kali 12208 7z scalpel_output
-rw-r--r-- 1 kali kali 12208 7z scalpel_output2
-rw-r--r-- 1 kali kali 12208 7z scalpel_output_fixed
-rw-r--r-- 1 kali kali 12208 7z scalpel_output_lab3
-rw-r--r-- 1 kali kali 12208 7z scalpel_output_lab3_clean
-rw-r--r-- 1 kali kali 12208 7z scalpel_output_lab3_clean5
-rw-r--r-- 1 kali kali 12208 7z scalpel_output_lab3_final
-rw-r--r-- 1 kali kali 12208 7z scalpel_output_lab3_fixed
-rw-r--r-- 1 kali kali 12208 7z scalpel_output_lab3_new
-rw-r--r-- 1 kali kali 12208 7z scalpel_output_lab4
-rw-r--r-- 1 kali kali 12208 7z scalpel_output_new
-rw-r--r-- 1 kali kali 12208 7z working_file
-rw-r--r-- 1 kali kali 12208 7z allowed.pcapng
-rw-r--r-- 1 kali kali 12208 7z blocked.pcapng
-rw-r--r-- 1 kali kali 12208 7z 'C:\InChap01 (1).dd'
-rw-r--r-- 1 kali kali 12208 7z dd_block_size_test_result.txt
-rw-r--r-- 1 kali kali 12208 7z dd_obs_testfile
-rw-r--r-- 1 kali kali 12208 7z dd_obs_test.sh
-rw-r--r-- 1 kali kali 12208 7z evidence_test.txt
-rw-r--r-- 1 kali kali 12208 7z file1.txt
-rw-r--r-- 1 kali kali 12208 7z file2.txt
-rw-r--r-- 1 kali kali 12208 7z file.txt
-rw-r--r-- 1 kali kali 12208 7z file.xz
-rw-r--r-- 1 kali kali 12208 7z firstnamelastnameEvidence_received.txt
-rw-r--r-- 1 kali kali 12208 7z hello_with_newline.txt
-rw-r--r-- 1 kali kali 12208 7z hello_without_newline.txt
-rw-r--r-- 1 kali kali 12208 7z ImoruMayowaPeterEvidence_received.txt
-rw-r--r-- 1 kali kali 12208 7z 2_lab_img_hex.txt
```

Decompressed derivative hash (working_file.pcap)

```
kali@kali: ~/Desktop
Session Actions Edit View Help
2_lab_img_hex.txt working_file
2_lab_img.jpg
(kali@kali)~/Desktop
$ sha256sum working_file.pcap
sha256sum: working_file.pcap: No such file or directory

(kali@kali)~/Desktop
$ ls -l ~/Desktop
-rw-r--r-- 1 kali kali 12208 7z 2_lab_img_recovered.jpg
-rw-r--r-- 1 kali kali 12208 7z 2_lab_img_strings.txt
-rw-r--r-- 1 kali kali 12208 7z lab2a_sys_summary.sh
-rw-r--r-- 1 kali kali 12208 7z lab2a_sys_summary.sh
-rw-r--r-- 1 kali kali 12208 7z lab2a
-rw-r--r-- 1 kali kali 12208 7z 'mkdir_Desktop_abaa'
-rw-r--r-- 1 kali kali 12208 7z ruleset.txt
-rw-r--r-- 1 kali kali 12208 7z scalpel_output
-rw-r--r-- 1 kali kali 12208 7z scalpel_output2
-rw-r--r-- 1 kali kali 12208 7z scalpel_output_fixed
-rw-r--r-- 1 kali kali 12208 7z scalpel_output_lab3
-rw-r--r-- 1 kali kali 12208 7z scalpel_output_lab3_clean
-rw-r--r-- 1 kali kali 12208 7z scalpel_output_lab3_clean5
-rw-r--r-- 1 kali kali 12208 7z scalpel_output_lab3_final
-rw-r--r-- 1 kali kali 12208 7z scalpel_output_lab3_fixed
-rw-r--r-- 1 kali kali 12208 7z scalpel_output_lab3_new
-rw-r--r-- 1 kali kali 12208 7z scalpel_output_lab4
-rw-r--r-- 1 kali kali 12208 7z scalpel_output_new
-rw-r--r-- 1 kali kali 12208 7z working_file
-rw-r--r-- 1 kali kali 12208 7z allowed.pcapng
-rw-r--r-- 1 kali kali 12208 7z blocked.pcapng
-rw-r--r-- 1 kali kali 12208 7z 'C:\InChap01 (1).dd'
-rw-r--r-- 1 kali kali 12208 7z dd_block_size_test_result.txt
-rw-r--r-- 1 kali kali 12208 7z dd_obs_testfile
-rw-r--r-- 1 kali kali 12208 7z dd_obs_test.sh
-rw-r--r-- 1 kali kali 12208 7z evidence_test.txt
-rw-r--r-- 1 kali kali 12208 7z file1.txt
-rw-r--r-- 1 kali kali 12208 7z file2.txt
-rw-r--r-- 1 kali kali 12208 7z file.txt
-rw-r--r-- 1 kali kali 12208 7z file.xz
-rw-r--r-- 1 kali kali 12208 7z firstnamelastnameEvidence_received.txt
-rw-r--r-- 1 kali kali 12208 7z hello_with_newline.txt
-rw-r--r-- 1 kali kali 12208 7z hello_without_newline.txt
-rw-r--r-- 1 kali kali 12208 7z ImoruMayowaPeterEvidence_received.txt
-rw-r--r-- 1 kali kali 12208 7z 2_lab_img_hex.txt
-rw-r--r-- 1 kali kali 12208 7z 2_lab_img.jpg

(kali@kali)~/Desktop
$ mv working_file working_file.pcap

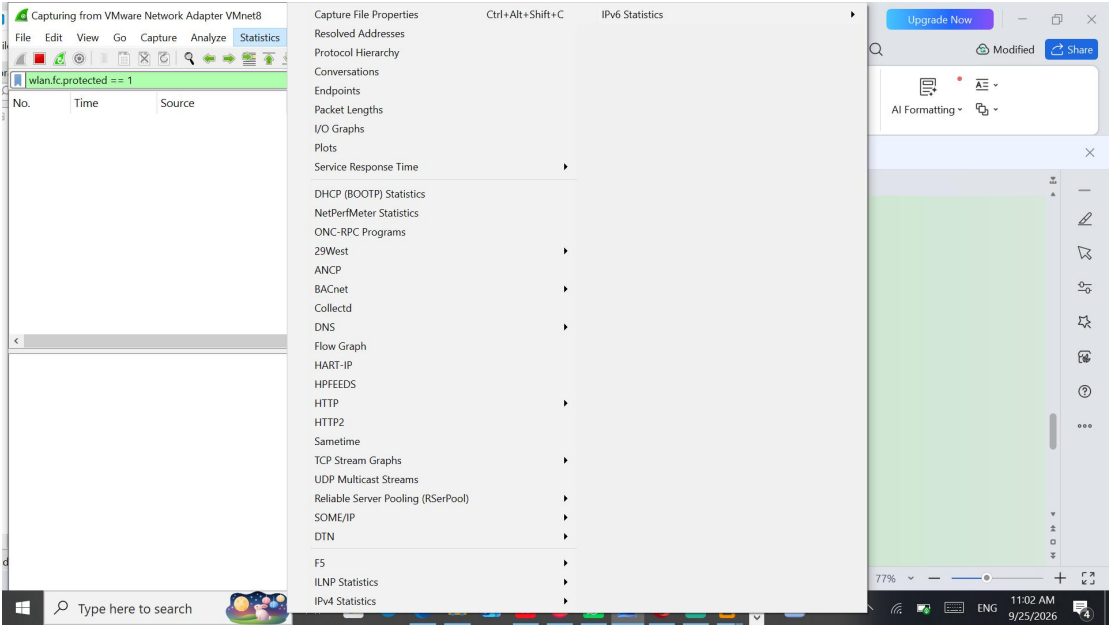
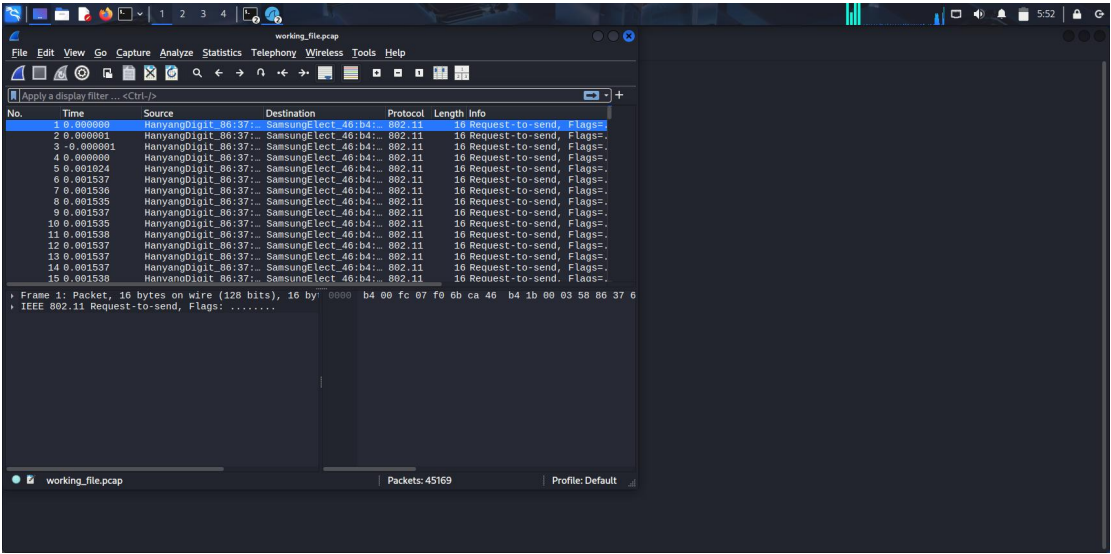
(kali@kali)~/Desktop
$ sha256sum working_file.pcap
c17a3f9b959e84f5befd476dbd55c67286d1e3ee9ab402d539cac0874ebb2d  working_file.pcap

(kali@kali)~/Desktop
$
```

REPORT

The analysis showed repeated IV values, a critical weakness in WEP. IV reuse allows attackers to recover keys and decrypt traffic, proving WEPs insecurity

EVIDENCE 3



REPORT

The analysis showed repeated IV values, a critical weakness in WEP. IV reuse allows attackers to recover keys and decrypt traffic, proving WEP's insecurity

EVIDENCE 4

```
kali@kali: ~/Desktop
Session Actions Edit View Help
~$ ls
120M 120M.7z allowed.pcapng blocked.pcapng 'ChalInChap01 (1).dd' dd_block_size_test_result.txt dd_obs_testfile dd_obs_test.sh evidence_test.txt file1.txt file2.txt file.txt file.xz FirstnameLastnameEvidence_received.txt hello_with_newline.txt hello_without_newline.txt ImoruMayowaPeterEvidence_received.txt j_lab_low_hex.txt j_lab_low.jpg j_lab_low_recovered.jpg j_lab_low_strings.txt lab2a_sys_summary.sh lab2b_sys_summary.sh Lab2a 'mdir Desktop_sb4a' ruleset.txt scalpel_output scalpel_output2 scalpel_output_fixed scalpel_output_lab3 scalpel_output_lab3_clean scalpel_output_lab3_clean5 scalpel_output_lab3_final scalpel_output_lab3_fixed scalpel_output_lab3_new scalpel_output_lab4 scalpel_output_new working_file.pcap

(kali@kali)~(~/Desktop)
$ aircrack-ng -w 1234567890 working_file.pcap
Total number of stations seen 10
Total number of packets read 45169
Total number of WEP data packets 15477
Total number of WPA data packets 0
Number of plaintext data packets 0
Number of decrypted WEP packets 0
Number of corrupted WEP packets 0
Number of decrypted WPA packets 0
Number of bad TKIP (WPA) packets 0
Number of bad COMP (WPA) packets 0

(kali@kali)~(~/Desktop)
$ ls -l working_file-dec.pcap
-rw-rw-r-- 1 kali kali 24 Sep 25 06:11 working_file-dec.pcap

(kali@kali)~(~/Desktop)
$
```

Statistics → Endpoints.

```
kali@kali: ~/Desktop
Session Actions Edit View Help
file1.txt scalpel_output_lab3
file2.txt scalpel_output_lab3_clean
file.txt scalpel_output_lab3_clean5
file.xz scalpel_output_lab3_clean5
file.xz scalpel_output_lab3_final
FirstnameLastnameEvidence_received.txt scalpel_output_lab3_fixed
hello_with_newline.txt scalpel_output_lab3_new
hello_without_newline.txt scalpel_output_lab4
ImoruMayowaPeterEvidence_received.txt scalpel_output_new
j_lab_low_hex.txt working_file.pcap
j_lab_low.jpg
j_lab_low_recovered.jpg
j_lab_low_strings.txt

(kali@kali)~(~/Desktop)
$ aircrack-ng -w 1234567890 working_file.pcap
Total number of stations seen 10
Total number of packets read 45169
Total number of WEP data packets 15477
Total number of WPA data packets 0
Number of plaintext data packets 0
Number of decrypted WEP packets 0
Number of corrupted WEP packets 0
Number of decrypted WPA packets 0
Number of bad TKIP (WPA) packets 0
Number of bad COMP (WPA) packets 0

(kali@kali)~(~/Desktop)
$ ls -l working_file-dec.pcap
-rw-rw-r-- 1 kali kali 24 Sep 25 06:11 working_file-dec.pcap

(kali@kali)~(~/Desktop)
$ tshark -r working_file-dec.pcap -q -z endpoints,ip
Warning: program compiled against libxml 215 using older 214

IPv4 Endpoints
Filter: <No Filter>
| Packets | | Bytes | | Tx Packets | | Tx Bytes | | Rx Packets | | Rx Bytes |
|
(kali@kali)~(~/Desktop)
$
```

Statistics → Conversations.

```
kali@kali: ~/Desktop
Session Actions Edit View Help

Total number of stations seen      10
Total number of packets read      45169
Total number of WEP data packets  15477
Total number of WPA data packets  0
Number of plaintext data packets  0
Number of decrypted WEP packets   0
Number of corrupted WEP packets   0
Number of decrypted WPA packets   0
Number of bad TKIP (WPA) packets  0
Number of bad COMP (WPA) packets  0

(kali@kali)~/Desktop
$ ls -l working_file-dec.pcap
-rw-rw-r-- 1 kali kali 24 Sep 25 06:11 working_file-dec.pcap

(kali@kali)~/Desktop
$ tshark -r working_file-dec.pcap -q -z endpoints,ip
Warning: program compiled against libxml 215 using older 214

IPv4 Endpoints
Filter:<No Filter>

+-----+-----+-----+-----+-----+-----+
| Packets | | Bytes | | Tx Packets | | Tx Bytes | | Rx Packets | | Rx Bytes |
+-----+-----+-----+-----+-----+-----+

(kali@kali)~/Desktop
$ tshark -r working_file-dec.pcap -q -z conv,ip
Warning: program compiled against libxml 215 using older 214

IPv4 Conversations
Filter:<No Filter>

+-----+-----+-----+-----+-----+-----+-----+-----+
| Relative | Duration | | | | | | | | Total | |
| Start | | | | | | | | | | | |
+-----+-----+-----+-----+-----+-----+-----+-----+

(kali@kali)~/Desktop
$
```

HARSH

```
kali@kali: ~/Desktop
Session Actions Edit View Help

(kali@kali)~/Desktop
$ tshark -r working_file-dec.pcap -q -z conv,ip
Warning: program compiled against libxml 215 using older 214

IPv4 Conversations
Filter:<No Filter>

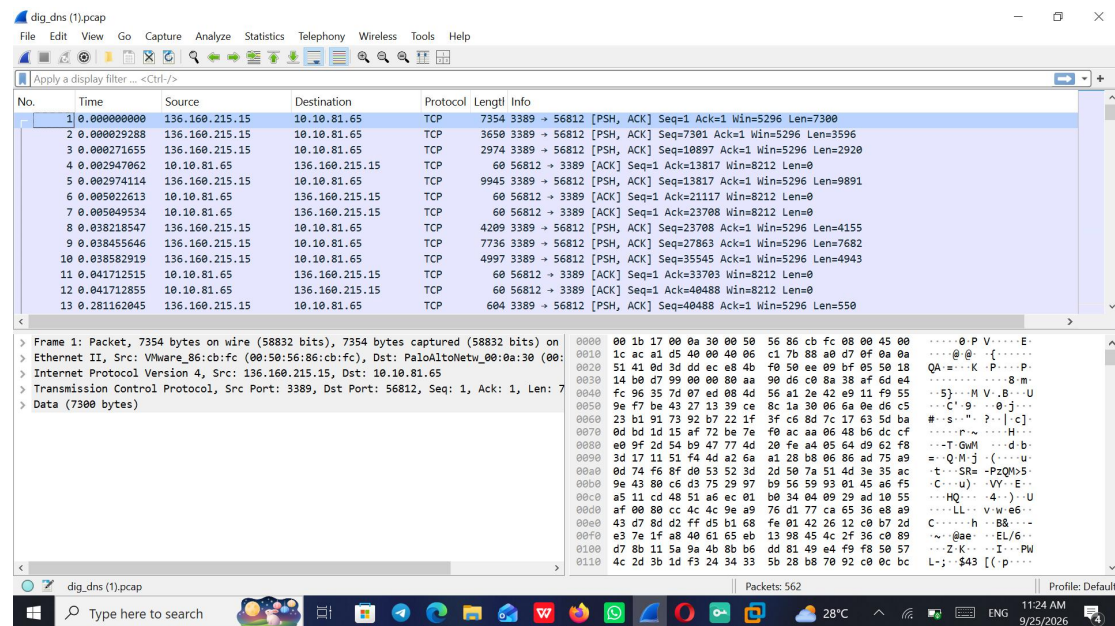
+-----+-----+-----+-----+-----+-----+-----+-----+
| Relative | Duration | | | | | | | | Total | |
| Start | | | | | | | | | | | |
+-----+-----+-----+-----+-----+-----+-----+-----+

(kali@kali)~/Desktop
$
```

REPORT

Decryption exposed endpoints and conversations, revealing communication between devices. Hashing the decrypted file confirmed forensic integrity. This shows that once WEP traffic is decrypted, sensitive information becomes visible.

EVIDENCE 5



REPORT

Wireshark's Export Objects feature was used to attempt file recovery. No recoverable objects were found in this capture, but the process demonstrates how attackers can reconstruct files from insecure WEP traffic. WPA2/WPA3 prevent this exposure.

EVIDENCE SIX


```
kali@kali: ~/Desktop
Session Actions Edit View Help

(kali@kali)~/Desktop
$ tshark -r working_file-dec.pcap -q -z conv,ip
Warning: program compiled against libxml 215 using older 214

IPv4 Conversations
Filter: <no filter>

```

Relative	Duration		←		→		Total	
Start			Frames	Bytes	Frames	Bytes	Frames	Bytes

```

(kali@kali)~/Desktop
$ sha256sum recovered_file.jpg
sha256sum: recovered_file.jpg: No such file or directory

(kali@kali)~/Desktop
$ sha256sum recovered_file.jpg
sha256sum: recovered_file.jpg: No such file or directory

(kali@kali)~/Desktop
$ sha256sum working_file.pcap
sha256sum working_file-dec.pcap
sha256sum recovered_file.jpg # if any
c17a3f9b95e84f5befd476dbd55c67286d1e3eea9ab402d5359cac0874ebb2d working_file.pcap
acc3386d8c8bc0b2d229281130b1899bfc81d70fdada5c34b3236c628f739c8 working_file-dec.pcap
sha256sum: recovered_file.jpg: No such file or directory

(kali@kali)~/Desktop
$
```

REPORT

SHA256 hashing verified the integrity of all files, ensuring chain of custody. This step proves that evidence was not altered during analysis.

EVIDENCE SEVEN

```
kali@kali: ~/Desktop
Session Actions Edit View Help

(kali@kali)~/Desktop
$ tshark -r working_file-dec.pcap -q -z conv,ip
Warning: program compiled against libxml 215 using older 214

IPv4 Conversations
Filter: <no filter>

```

Relative	Duration		←		→		Total	
Start			Frames	Bytes	Frames	Bytes	Frames	Bytes

```

(kali@kali)~/Desktop
$ sha256sum recovered_file.jpg
sha256sum: recovered_file.jpg: No such file or directory

(kali@kali)~/Desktop
$ sha256sum recovered_file.jpg
sha256sum: recovered_file.jpg: No such file or directory

(kali@kali)~/Desktop
$ sha256sum working_file.pcap
sha256sum working_file-dec.pcap
sha256sum recovered_file.jpg # if any
c17a3f9b95e84f5befd476dbd55c67286d1e3eea9ab402d5359cac0874ebb2d working_file.pcap
acc3386d8c8bc0b2d229281130b1899bfc81d70fdada5c34b3236c628f739c8 working_file-dec.pcap
sha256sum: recovered_file.jpg: No such file or directory

(kali@kali)~/Desktop
$
```

REPORT

EVIDENEC EIGHT

```

kali@kali: ~/Desktop
Session Actions Edit View Help

kali@kali:~/Desktop
$ tsshark -r working_file-dec.pcap -q -z conv,ip

Warning: program compiled against libxml 215 using older 214

IPv4 Conversations
Filter: (No Filter)

```

Relative	Duration		←		→		Total
Start			Frames	Bytes	Frames	Bytes	Frames
<pre> kali@kali:~/Desktop \$ sha256sum recovered_file.jpg sha256sum: recovered_file.jpg: No such file or directory kali@kali:~/Desktop \$ sha256sum recovered_file.jpg sha256sum: recovered_file.jpg: No such file or directory kali@kali:~/Desktop \$ sha256sum working_file.pcap sha256sum working_file-dec.pcap sha256sum recovered_file.jpg # if any c17a3f9b935e84f5befd676dbd55c67286d1e3ee9ab402d5359cac087aebb2d working_file.pcap acc330668c8b0eb2d29281130b18990f8c81d70fda5c34b3236c620f7739c8 working_file-dec.pcap sha256sum: recovered_file.jpg: No such file or directory kali@kali:~/Desktop \$</pre>							

Evidence 8 represents the final consolidation of Lab 9. All evidences were compiled into a structured forensic report. The lab confirmed WEP's insecurity and reinforced the importance of WPA2/WPA3 encryption.